Aaron Ghosh

647-675-3762 | aaronghosh9@gmail.com | linkedin.com/in/aaron-ghosh | github.com/aaronghsh

Education

McMaster University **Expected Graduation April 2028**
Bachelor of Engineering in Electrical Engineering (B.Eng.) (Co-op) *Hamilton, Ontario*

Relevant Coursework: Logic Design, Signals & Systems, Microprocessor Systems, Circuits & Systems, Data Structures & Algorithms, Electrical Systems Integration, Communication Systems, Statistics

Skills

Languages: Python, Perl, C, C++, ARM Assembly, SystemVerilog, Verilog, MATLAB, Bash
Tools & Platforms: ROS, Linux, Cadence OrCAD PSpice, Git, GitHub, Intel Quartus, ModelSim, Altium Design
Hardware & Design: RTL Design, FPGA Design, Embedded/Real-Time Systems, Timing Analysis

Experience

General Dynamics Mission Systems – Canada **May 2026 – April 2027**
Systems Engineering Co-op *Ottawa, Ontario*

- Perform system integration and verification across Windows and RHEL servers, validating application deployment, network communication, and system configuration across virtualized environments
- Investigate system-level defects through logs and configuration analysis to isolate faults across operating system, networking, runtime, and application layers.
- Develop and maintain Python and PowerShell scripts for system health checks, configuration, and application auditing, leveraging AI-assisted tools to automate test case verification workflows.

Supply & Demand **May 2025 – August 2025**
Experience Technician *Mississauga, Ontario*

- Assisted with system calibration, fault diagnosis, and verification of projection-mapping systems using OpenCV and LiDAR-based motion sensors to ensure precise alignment and real-time collision detection
- Integrated AR subsystems with physical display modules, synchronizing virtual overlays to user motion data for a seamless mixed-reality experience
- Diagnosed and resolved performance issues across hardware and network systems, reducing system latency while improving accuracy and reliability

Projects

Autonomous Electric Vehicle: *Python, C++, ROS, Linux, Optimization, Embedded Systems* **April 2026**

- Developed **embedded software** in C++ and Python using **ROS on an ARM-based NVIDIA Jetson Nano** for real-time autonomous vehicle control
- Processed and fused **LiDAR, camera, IMU, and odometry data** using a **probabilistic occupancy grid mapper** with **inverse sensor models and log-odds updates** for real-time state mapping
- Designed and validated **feedback-linearizing** and **PD control algorithms** using **QP-based virtual barriers** for real-time collision avoidance and safe steering
- Integrated and debugged a **multi-node ROS system**, validating hardware-software integration and real-time performance using **Linux tools, ROS logging, and RViz-based simulation**

Configurable Memory Controller: *Perl, SystemVerilog, Intel Quartus, UVM Concepts* **November 2025**

- Implemented a parameterized memory controller in **SystemVerilog** supporting configurable data width, memory depth, read/write latency cycles, and burst transactions
- Designed **synthesizable RTL** with distinct **read/write FSMs** and ready/valid handshake protocols
- Developed a self-checking **SystemVerilog** testbench for functional verification using **UVM-inspired** transaction and scoreboard concepts to validate read/write functionality
- Automated verification with **Perl scripts** to parse simulation logs and generate CSV summaries of test results

3D LiDAR Scanner: *C, Python, Assembly, Open3D, Microcontroller, Embedded Systems* **April 2025**

- Designed and implemented an embedded system capable of 3D spatial mapping using a **TI-MSP432E401Y microcontroller**, **VL53L1X Time-of-Flight sensor**, and **28BYJ-48 stepper motor**
- Programmed embedded firmware in **C** and **Assembly**, interfacing the ToF sensor via **I2C** and transmitting sensor data over **UART** to a Python-based visualization script
- Processed sensor measurements in **Python** and rendered **interactive 3D point clouds** using **Open3D** to reconstruct scanned environments

CMOS XOR Gate: *MOSFETs, Digital Logic, Circuit Design, PSpice* **March 2025**

- Prototyped a **CMOS XOR logic gate** using discrete **NMOS** & **PMOS** transistors from CD4007B ICs
- Verified functionality using **logic analyzers** and static/dynamic waveform tests
- Measured **rise/fall times** and **propagation delay** to evaluate timing performance and validate XOR switching through **voltage-level** and waveform analysis